packed with pumice soaked in Conc. H_2SO_4 to absorb water (ii) a set of bulbs containing a strong solution of KOH to absorb CO_2 and finally (iii) a guard tube filled with anhydrous $CaCl_2$ to prevent the entry of moisture from atmosphere.

Procedure: The combustion tube is heated strongly to dry its content. It is then cooled slightly and connected to the absorption apparatus. The other end of the combustion tube is open for a while and the boat containing weighed organic substance is introduced. The tube is again heated strongly till the substance in the boat is burnt away. This takes about 2 hours. Finally, a strong current of oxygen is passed through the combustion tube to sweap away any traces of carbon dioxide or moisture which may be left in it. The U-tube and the potash bulbs are then detached and the increase in weight of each of them is determined.

Calculation:

Weight of the organic substance taken = w g

Increase in weight of H_2O = x g

Increase in weight of CO_2 = y g

18 g of H_2O contains 2g of hydrogen

$\therefore$ x g of H_2O contains $\left(\frac{2}{18}x\right)$g of hydrogen

Percentage of hydrogen=$\left(\frac{2}{18}\times\frac{x}{w}\times100\right)$%

44g of CO_2 contains 12g of carbon

$\therefore$ y g of CO_2 contains $\left(\frac{12}{44}y\right)$ g of carbon

Percentage of Carbon=$\left(\frac{12}{44}\times\frac{y}{w}\times100\right)$%

Note:

1. If the organic substance under investigation also contain N, it will produce oxides of nitrogen on combustion. A spiral of copper is introduced at the combustion tube, to reduce the oxides of nitrogen to nitrogen which escapes unabsorbed.

2. If the compound contains halogen as well, a spiral of silver is also introduced in the combustion tube. It converts halogen into silver halide.

3. In case if the substance also contains sulphur, the copper oxide in the combustion tube is replaced by lead chromate. The SO_2 formed during combustion is thus converted to lead sulphate and prevented from passing into the absorption unit.

Worked out example: 1

0.26g of an organic compound gave 0.039 g of water and 0.245 g of carbon dioxide on combustion. Calculate the percentage of C & H

Weight of organic compound = 0.26g

Weight of water = 0.039g

Weight of CO_2 = 0.245g

Percentage of hydrogen

18 g of water contain 2 g of hydrogen

0.039 g of water contain $\frac{2}{18}\times\frac{0.039}{0.26}$

% of hydrogen = $\frac{0.039}{0.26}\times\frac{2}{18}\times100$ =1.66%

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